Tutorial:
MDANSE basics

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MDANSE Workshop 2026

03/11/2026



Science and Technology Facilities Council
ISIS Neutron and Muon Source

Schedule: introduction

1. The origins of MDANSE.
2. The landscape of molecular dynamics analysis tools.
3. Design decisions: why do we “convert” a trajectory?
4. The relationship between MDANSE and MDANSE_GUI.

Schedule: practicals

1. Inspecting trajectory contents using MDANSE_GUI.
2. Running general-purpose analysis (PDF, DOS, etc.)
3. Saving and re-using scripts.
4. Isotropic and anisotropic systems.
5. Equilibration and simulation stability.

Software check

Please check that you have Python, any version from 3.11 to 3.14, and a virtual environment with:

- mdanse[all]
- mdanse_gui[all]
- ase
- asap3

If any of those are missing, start installing them now.

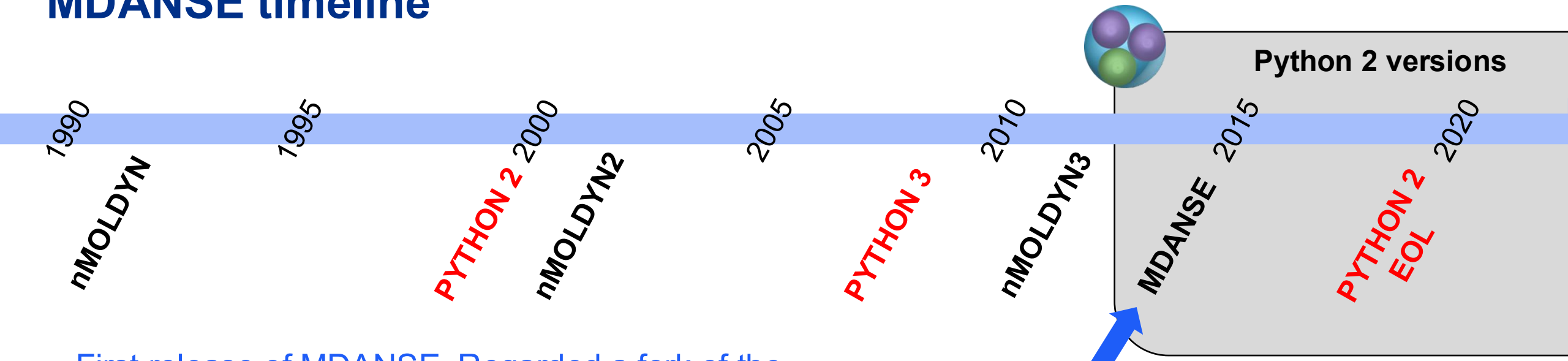
Also, once they are installed, please run:

```
mdanse_gui
```

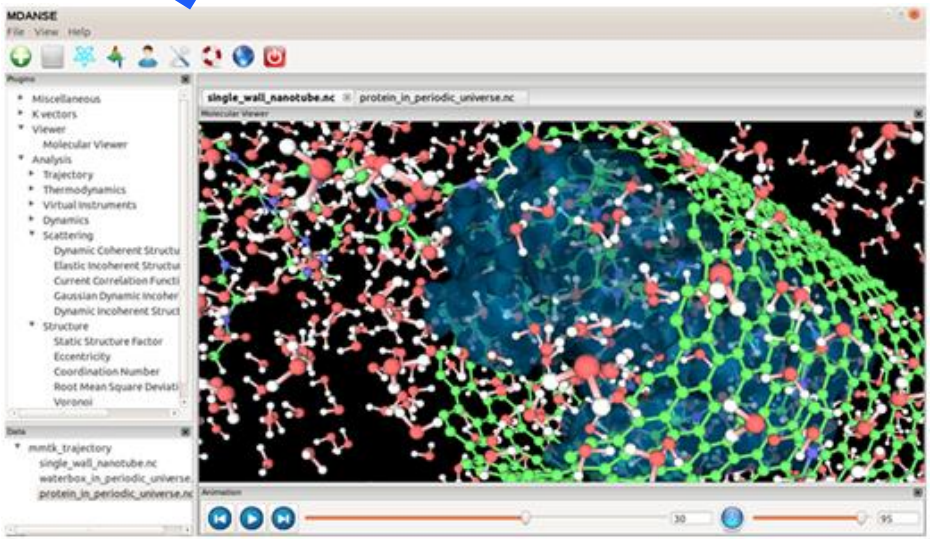
Note:
We are using Linux workspaces,
which means installing things is easy.

Windows users have told me
that they had problems
installing ASAP using pip,
as it needed a C++ compiler to build.

MDANSE timeline



- First release of MDANSE. Regarded a fork of the nMOLDYN3 program and incorporated code from ScientificPython and MMTK.*
- Not all features were retained in the transition from nMOLDYN to MDANSE.
- Originally developed at the ILL by E. Pellegrini et al.



JOURNAL OF CHEMICAL INFORMATION AND MODELING — Application Note — pubs.acs.org/jcim

MDANSE: An Interactive Analysis Environment for Molecular Dynamics Simulations

G. Goret, B. Aoun, and E. Pellegrini*

Computing for Science Group, Institut Laue-Langevin, 71 avenue des Martyrs, 38042 Grenoble, France

*https://github.com/ISISNeutronMuon/MDANSE/pull/169

MDANSE timeline

Python 2 versions

2021

2022

MDANSE 1.5.0

MDANSE 1.5.1

2023

MDANSE 1.5.2

2024

Python 3 versions

2025

MDANSE 2.0.0a1

MDANSE 2.0.0a2

MDANSE 2.0.0b1

2026

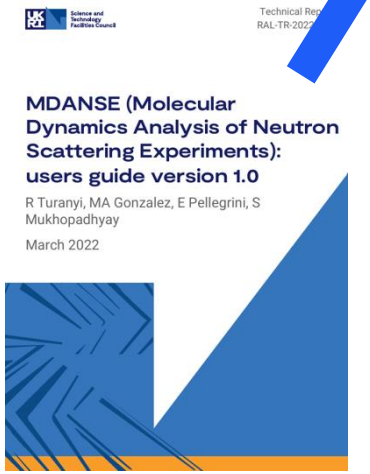
MDANSE 2.0.0

MDANSE 2.0.1

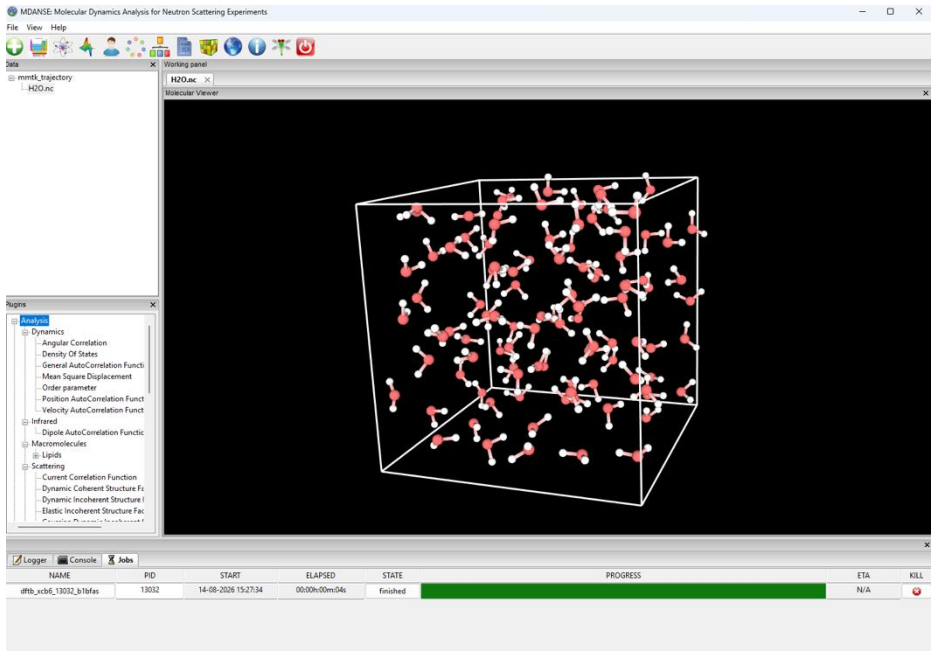
MDANSE 2.0.2

2027

MDANSE 2.x.y?



- A collaboration between software engineers and scientist at the ILL and ISIS to update MDANSE so that it can be installed and run on modern hardware.
- MDANSE repository moved to GitHub.
- MDANSE 1.5.2 was the last release which used Python 2.



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www.isis.stfc.ac.uk

@isisneutronmuon

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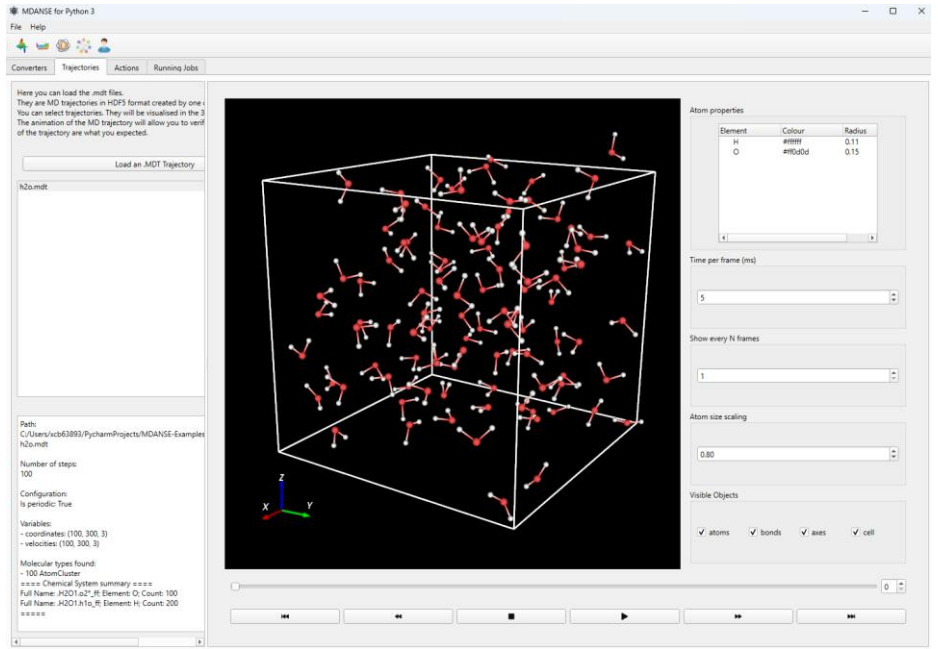
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MDANSE 2.0.0a1
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MDANSE 2.x.y?

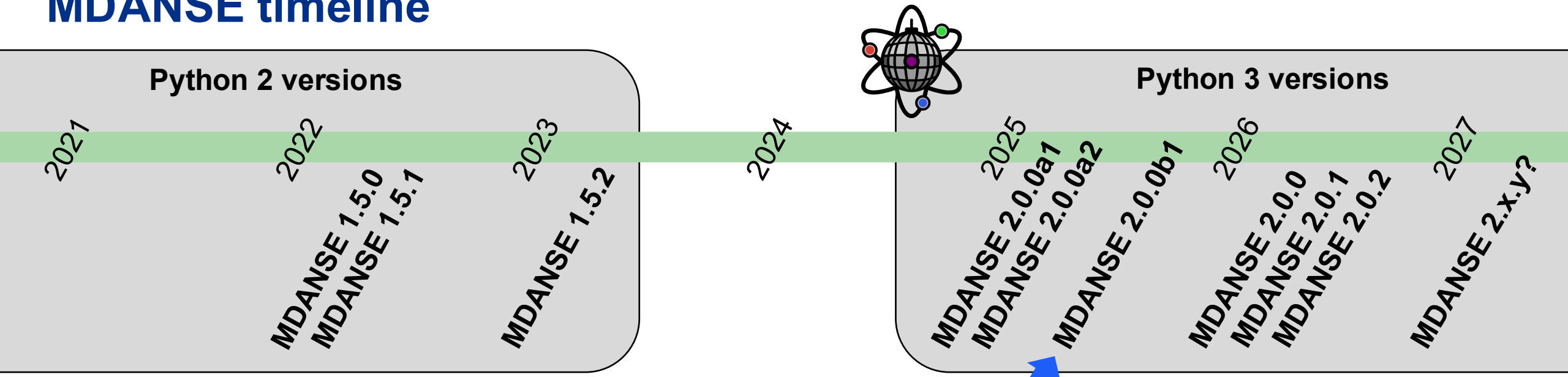
- First alpha release of MDANSE 2 and the first release to use Python 3 which also included a complete overhaul of the front end to use Qt and VTK.
- Second alpha release included new plotting tools.



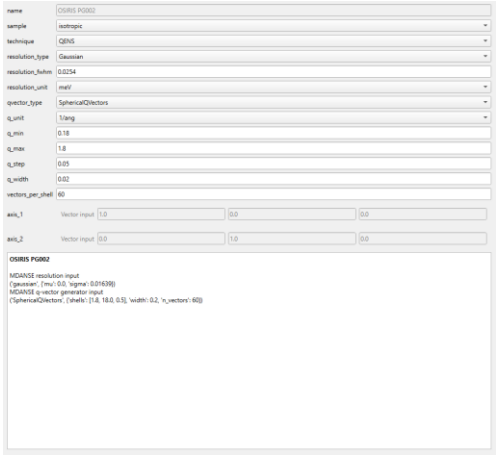
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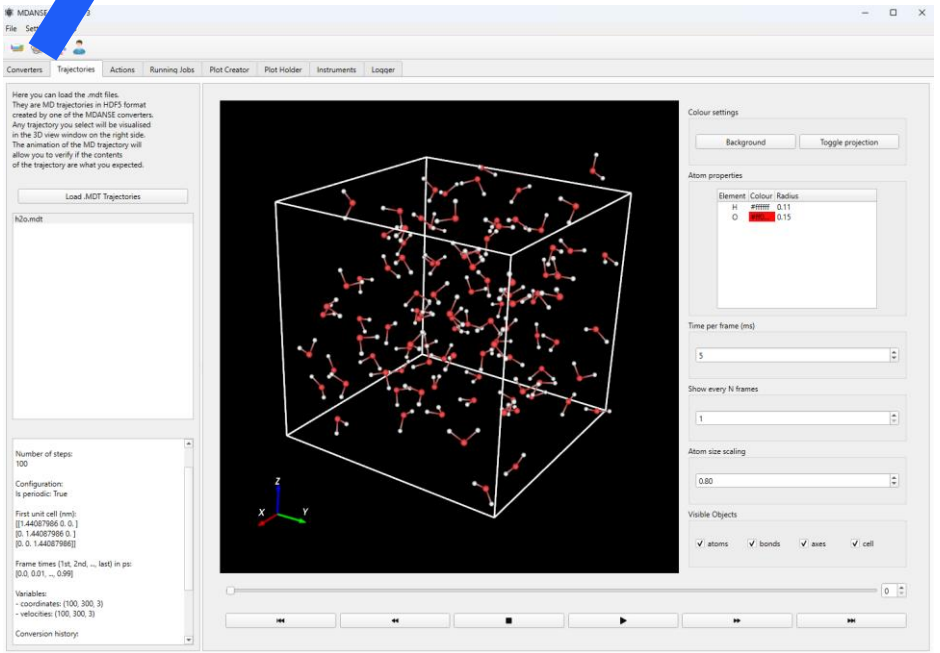
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MDANSE timeline



- First beta release of MDANSE which included further fixes and improvements to the converter and analysis code.
- New analysis calculations (van Hove and current-correlation functions) and a new instrument profiles settings were added.





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MDANSE 2.0.0

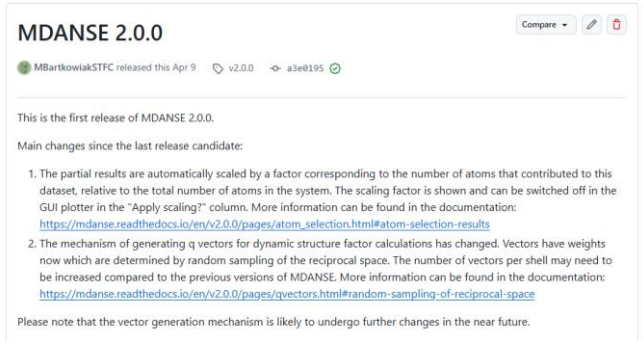
MDANSE 2.0.1

MDANSE 2.0.2

2027

MDANSE 2.x.y?

- First official MDANSE 2 release which included a huge number of fixes, improvements and new features over the beta version.
- 2.0.1 included some further updates and fixes.
- Manuscript in preparation.



MDANSE 2.0.0

Compare

MBartkowiakSTFC released this Apr 9 v2.0.0 a3e0195

This is the first release of MDANSE 2.0.0.

Main changes since the last release candidate:

1. The partial results are automatically scaled by a factor corresponding to the number of atoms that contributed to this dataset, relative to the total number of atoms in the system. The scaling factor is shown and can be switched off in the GUI plotter in the "Apply scaling?" column. More information can be found in the documentation: https://mdanse.readthedocs.io/en/v2.0.0/pages/atom_selection.html#atom-selection-results
2. The mechanism of generating q vectors for dynamic structure factor calculations has changed. Vectors have weights now which are determined by random sampling of the reciprocal space. The number of vectors per shell may need to be increased compared to the previous versions of MDANSE. More information can be found in the documentation: https://mdanse.readthedocs.io/en/v2.0.0/pages/qvectors.html#random-sampling-of-reciprocal-space

Please note that the vector generation mechanism is likely to undergo further changes in the near future.

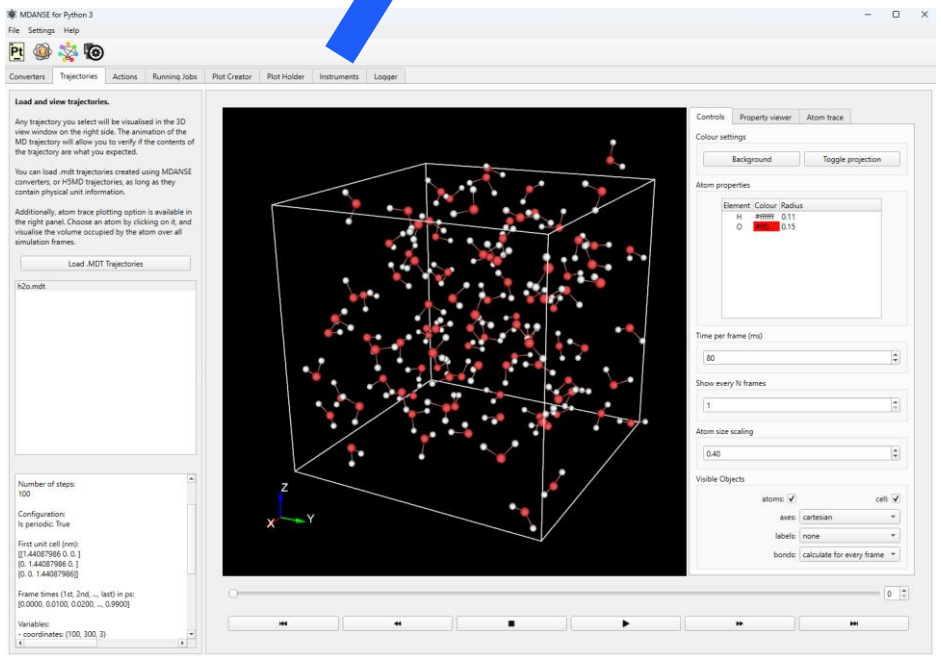


MDANSE 2.0.1 Latest

Compare

MBartkowiakSTFC released this Apr 30 v2.0.1 f443d54

MDANSE 2.0.1 improves the mechanism of q vector generation. Please refer to the documentation (https://mdanse.readthedocs.io/en/latest/pages/qvectors.html#random-sampling-of-reciprocal-space) for more details.



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Data check

In this tutorial we will need MD trajectories.

The plan is to generate them by running a script.

Start running the scripts called

`run_isotropic_trajectory.py`

`run_anisotropic_trajectory.py`

Molecular dynamics analysis tools – some alternatives

- There are many options to choose from.
- Pick your tools to match the problem you are solving.

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

Neutron-related

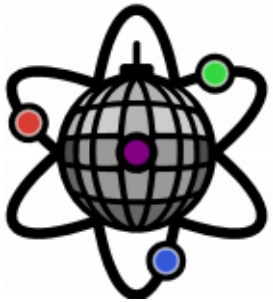




LiquidLib



dynasor



MDANSE

… OCLIMAX?

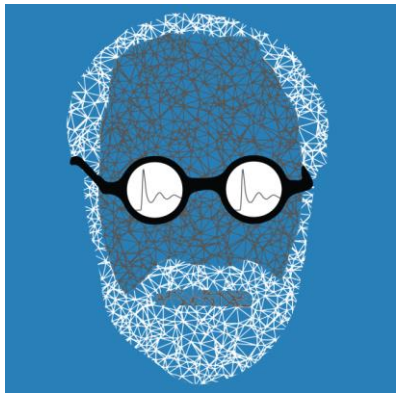
Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

General-purpose

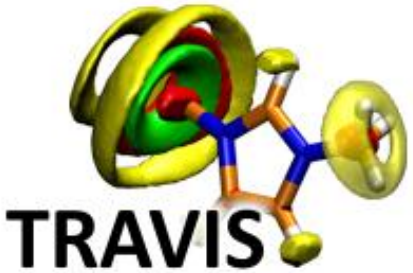
MD ANALYSIS

MDTRAJ



Freud

mdapy



VMD Visual Molecular Dynamics

MDANSE

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

For GROMACS

CHAPERONg



StreaMD

Molecular dynamics analysis tools – some alternatives

- There are a lot of tools that can be used to view and analyze your molecular dynamics trajectories. Some of these may be better or worse for the research problem that you are trying to solve.

Has a GUI



OVITO



Moltwister







Optimised for speed…?



MDANSE basic workflow

Trajectory conversion

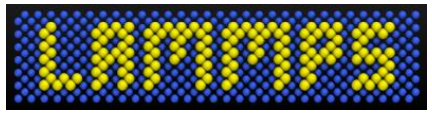












- Run a molecular dynamics simulation dump positions, velocity, etc.
- Convert trajectory to MDT (HDF5) format.

MDANSE basic workflow

Trajectory conversion

- Run a molecular dynamics simulation dump positions, velocity, etc.
- Convert trajectory to MDT (HDF5) format.

MD analysis



```
                self._nFrames,
            )
            self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] = np.zeros(
                self._nFrames,
            )
        return

    rho_l, rho_t = x
    n_configs = self.configuration["frames"]["n_configs"]
    for pair_str, (label_i, label_j) in self.labels:
        corr_l = correlate(
            rho_l[label_i], rho_l[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_l[label_i].shape[2])
        self._outputData[f"j(q,t)_long_{pair_str}"][index, :] += corr_l.real
        corr_t = correlate(
            rho_t[label_i], rho_t[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_t[label_i].shape[2])
        self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] += corr_t.real

def finalize(self):
    """Normalize, Fourier transform and write the results out."""
```

- Load trajectory and run analysis calculation.
- Save results as MDA (HDF5) or other formats.

MDANSE basic workflow

Trajectory conversion

- Run a molecular dynamics simulation dump positions, velocity, etc.
- Convert trajectory to MDT (HDF5) format.

MD analysis

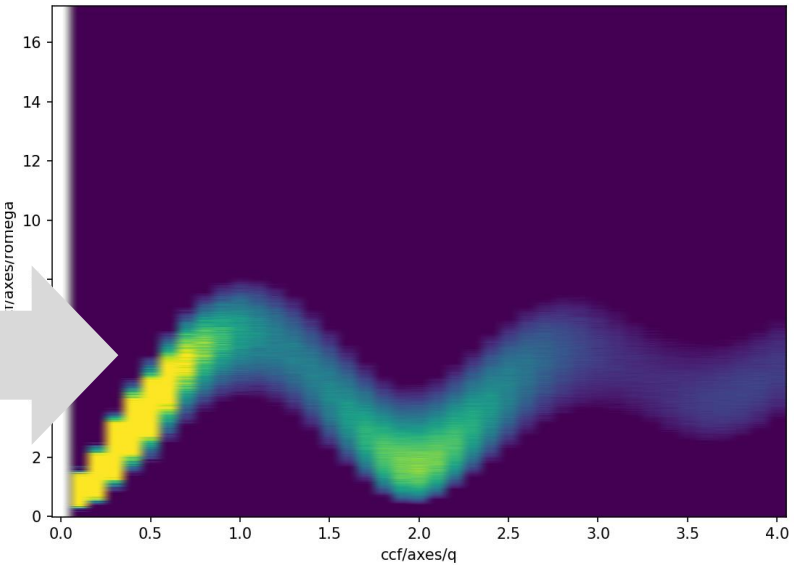
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        return

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    for pair_str, (label_i, label_j) in self.labels:
        corr_l = correlate(
            rho_l[label_i], rho_l[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_l[label_i].shape[2])
        self._outputData[f"j(q,t)_long_{pair_str}"][index, :] += corr_l.real
        corr_t = correlate(
            rho_t[label_i], rho_t[label_j][:n_configs], mode="valid"
        )[
            :,
            0,
            0,
        ] / (3 * n_configs * rho_t[label_i].shape[2])
        self._outputData[f"j(q,t)_trans_{pair_str}"][index, :] += corr_t.real

def finalize(self):
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- Load trajectory and run analysis calculation.
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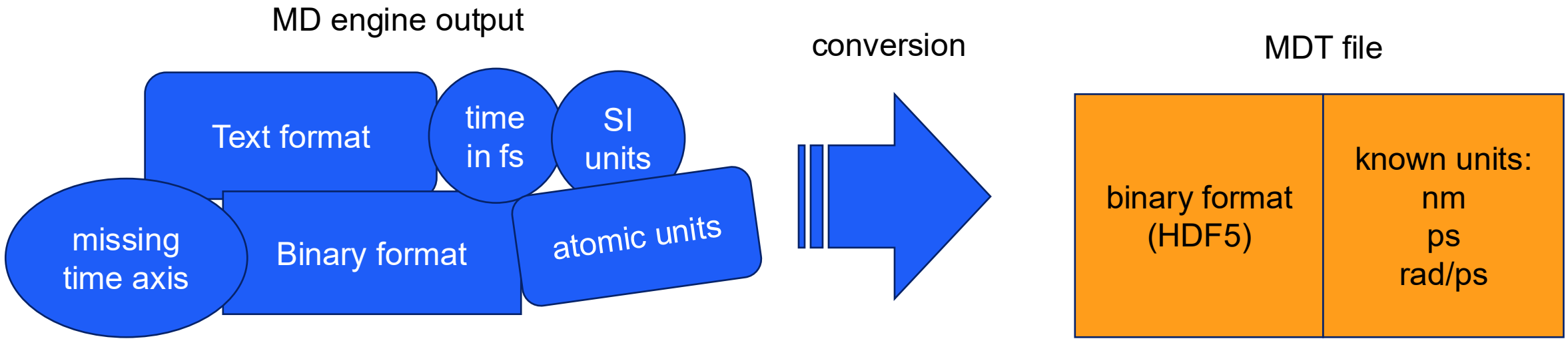
Plotting



- Load (MDA format) analysis results and plot using MDANSE GUI or with other tools/libraries.

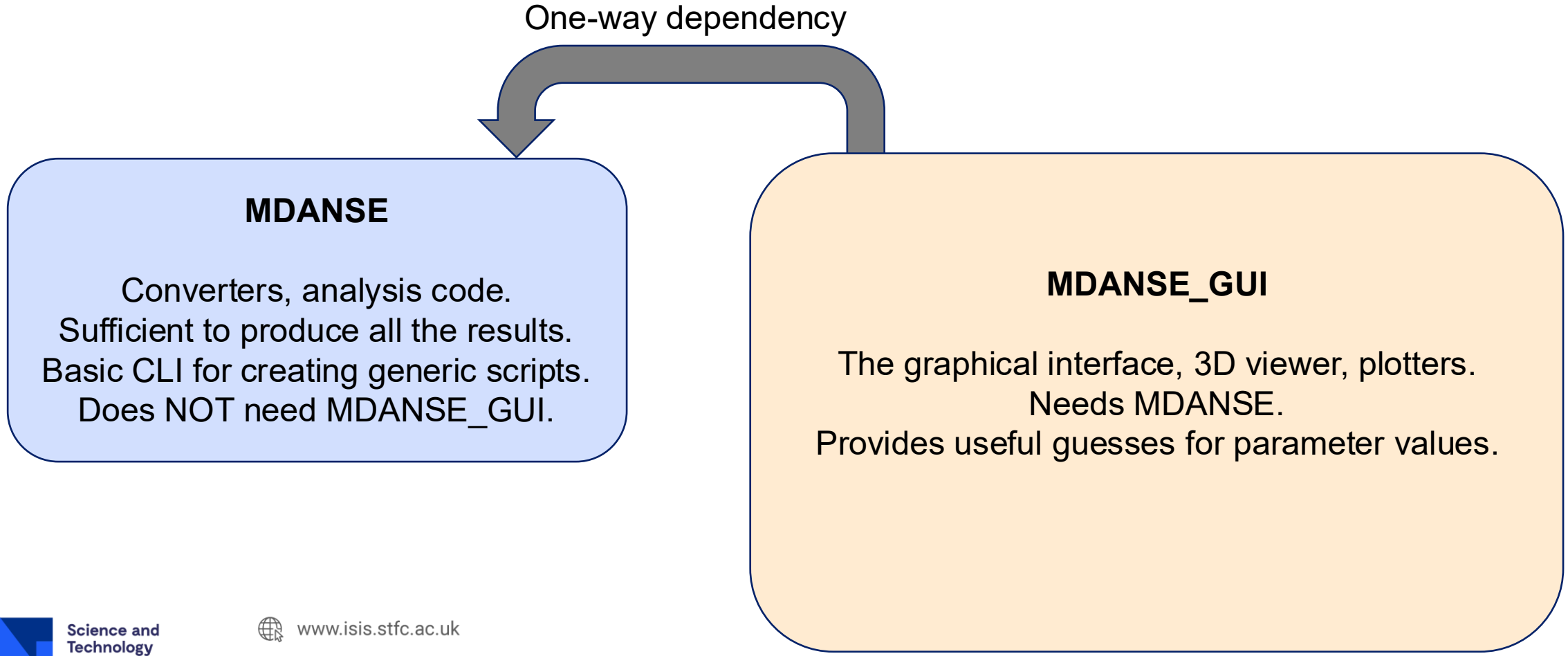
MDANSE converters

- A few general converters have been included, which use other MD analysis libraries to convert them to the MDANSE format (HDF).
- This should mean most MD topology and trajectory files can now be converted.



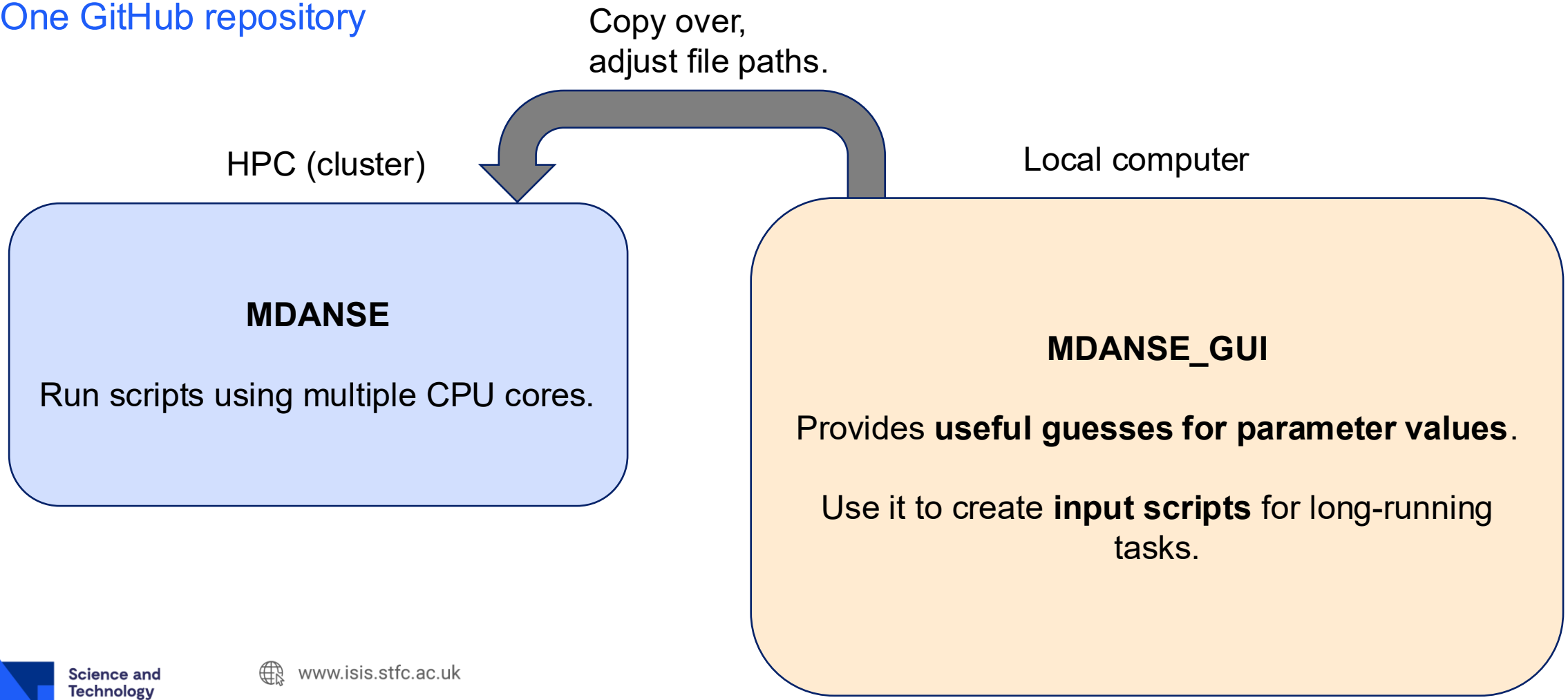
MDANSE and MDANSE_GUI

- Two Python packages
- One GitHub repository



MDANSE and MDANSE_GUI: a use case.

- Two Python packages
- One GitHub repository



MDANSE Basics: practicals.

We're about to start the GUI.

You're welcome to run

`mdanse_gui -h`

to find out that some things can be customised.

You can also run

`create_mdanse_link`

if you like having a desktop icon for the GUI.

You can just start the GUI by running

`mdanse_gui`

Trick question time

- We just used ASE to run MD simulations and write out trajectories. **Which converter** should we use to get the trajectories into MDANSE?
- The MDANSE GUI is divided into tabs. If you're looking for converters, **which is the right tab** to go to?



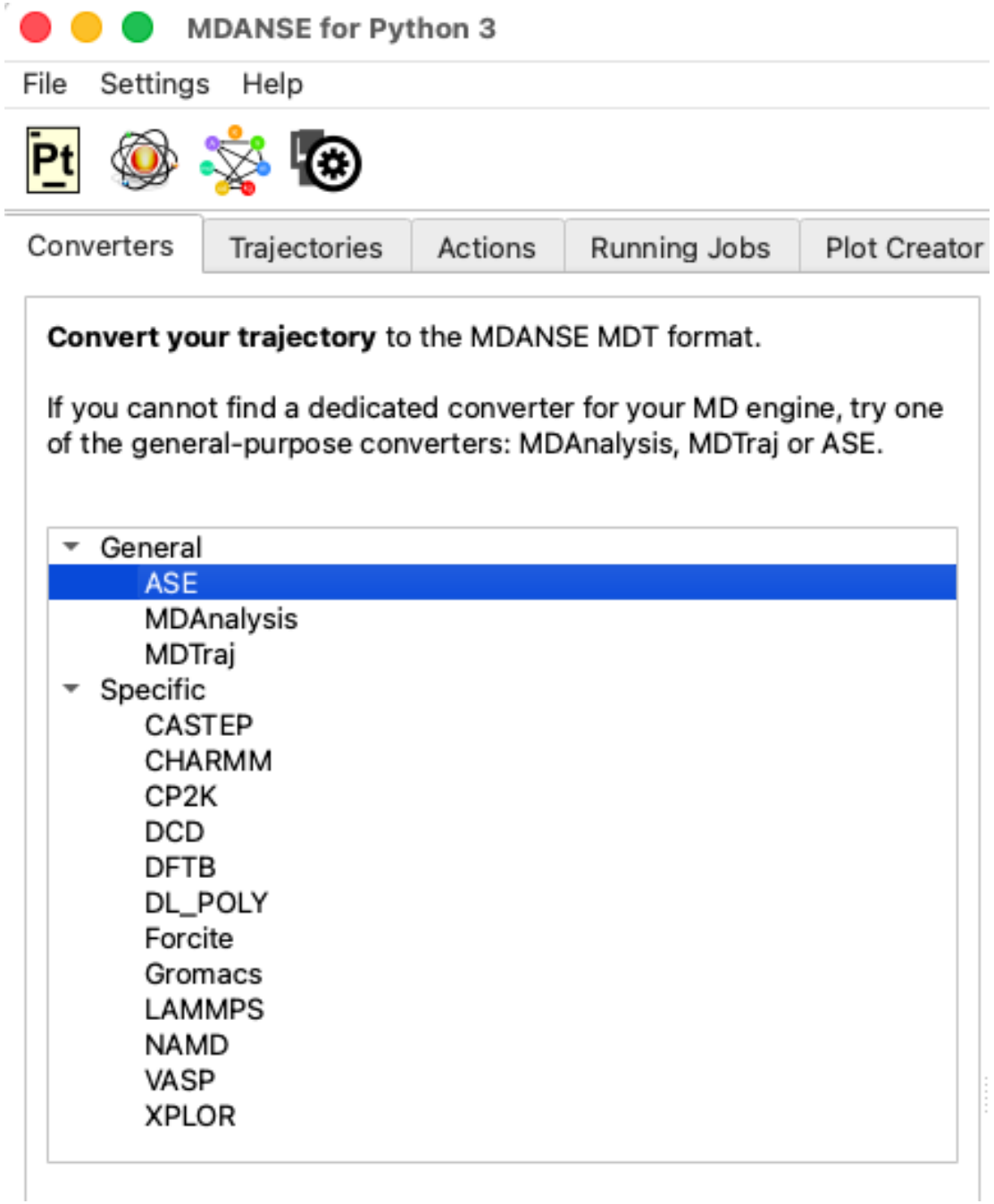
Trick question time

- We just used ASE to run MD simulations and write out trajectories. **Which converter** should we use to get the trajectories into MDANSE?

ASE converter

- The MDANSE GUI is divided into tabs. If you're looking for converters, **which is the right tab** to go to?

Converters tab, but we also accept the answer "the first one".

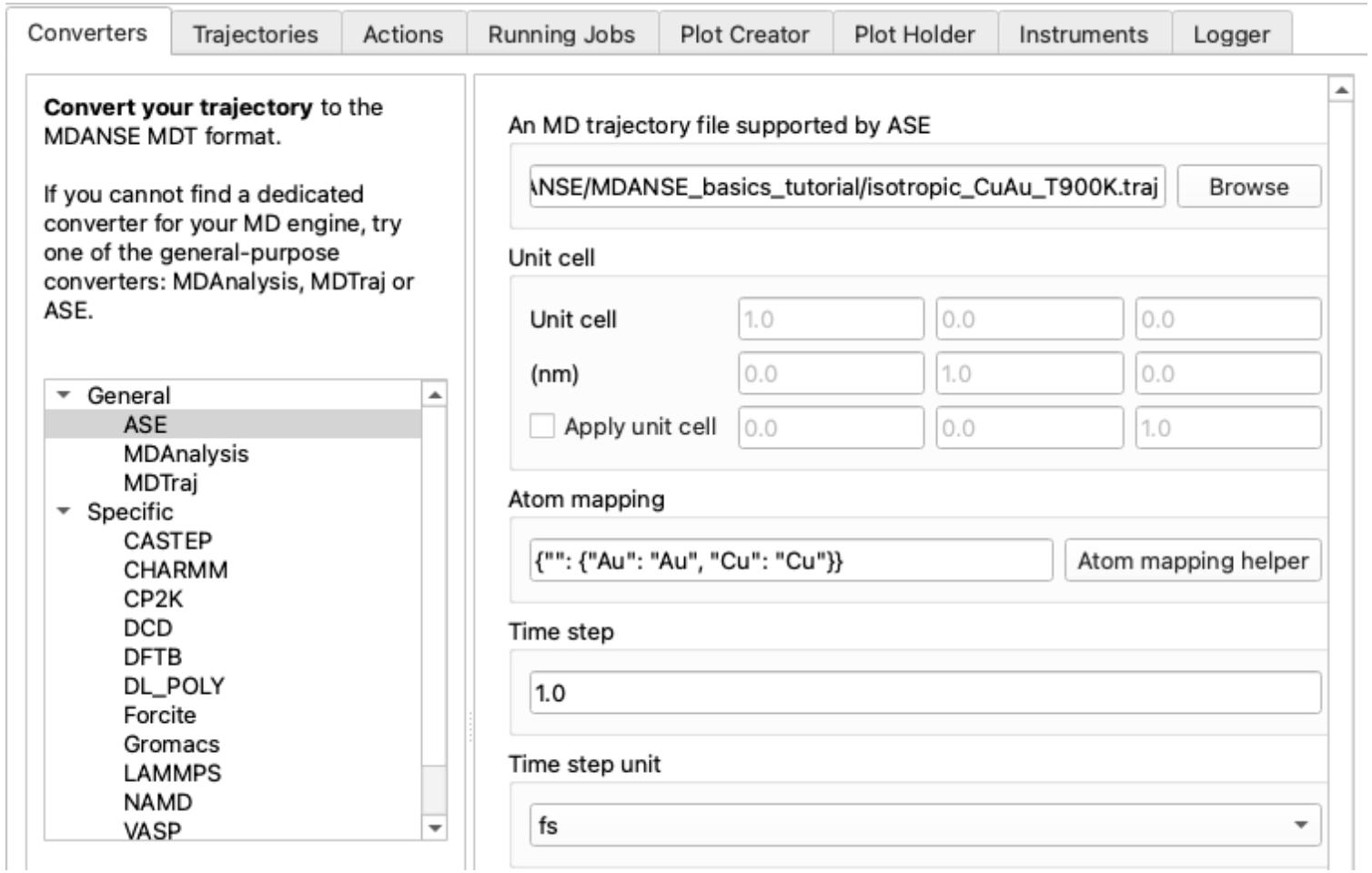


MDANSE Basics: practicals.

1. Convert both trajectories we just created.

- Check the input parameters. Some of them may need to be set manually to correct values.
- If you're not sure about some values, check the original script used for running the MD simulation.

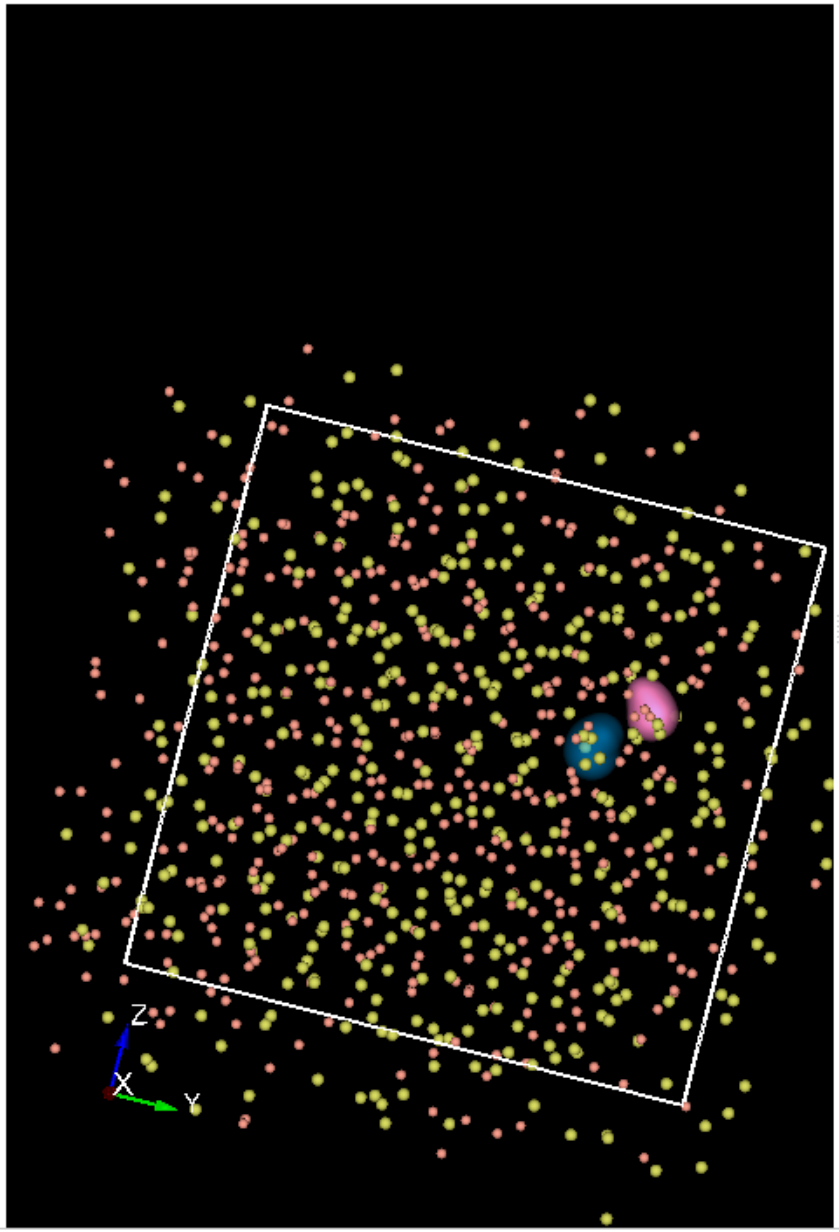
```
timestep=5.0 * units.fs,
```

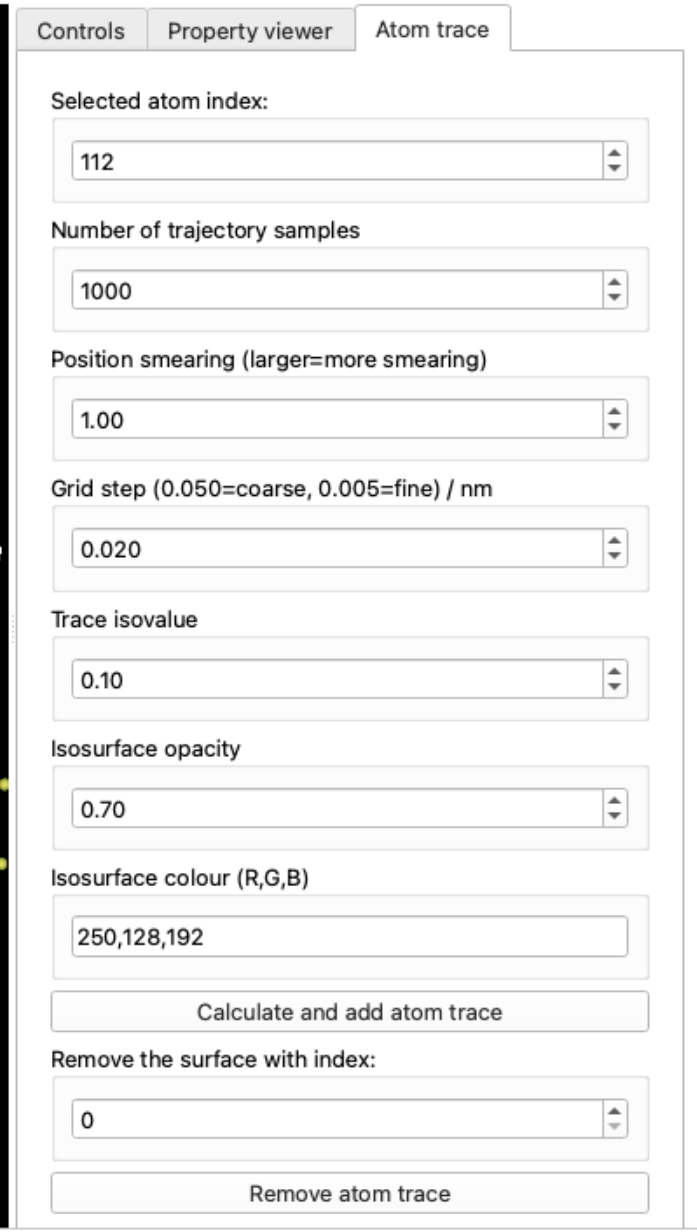


MDANSE Basics: Trajectory tab

Have a look at your trajectory:

- Switch off the bonds in the 3D view.
- Watch the animation of your trajectory.
- See how far an atom has moved: calculate the atom trace for some atoms.





MDANSE Basics: Trajectory tab

Find some numbers:

- How many atoms are in the trajectory?
- What are the first and last values on the time axis?
- What are the dimensions of the unit cell?
- What are the electric charge values on the atoms?

MDANSE Basics: GUI history

1. Switch off the GUI.
2. Start it again.
3. Go to File -> Open Recent Trajectories
4. Bring back the trajectories you were using before.

MDANSE Basics: GUI settings

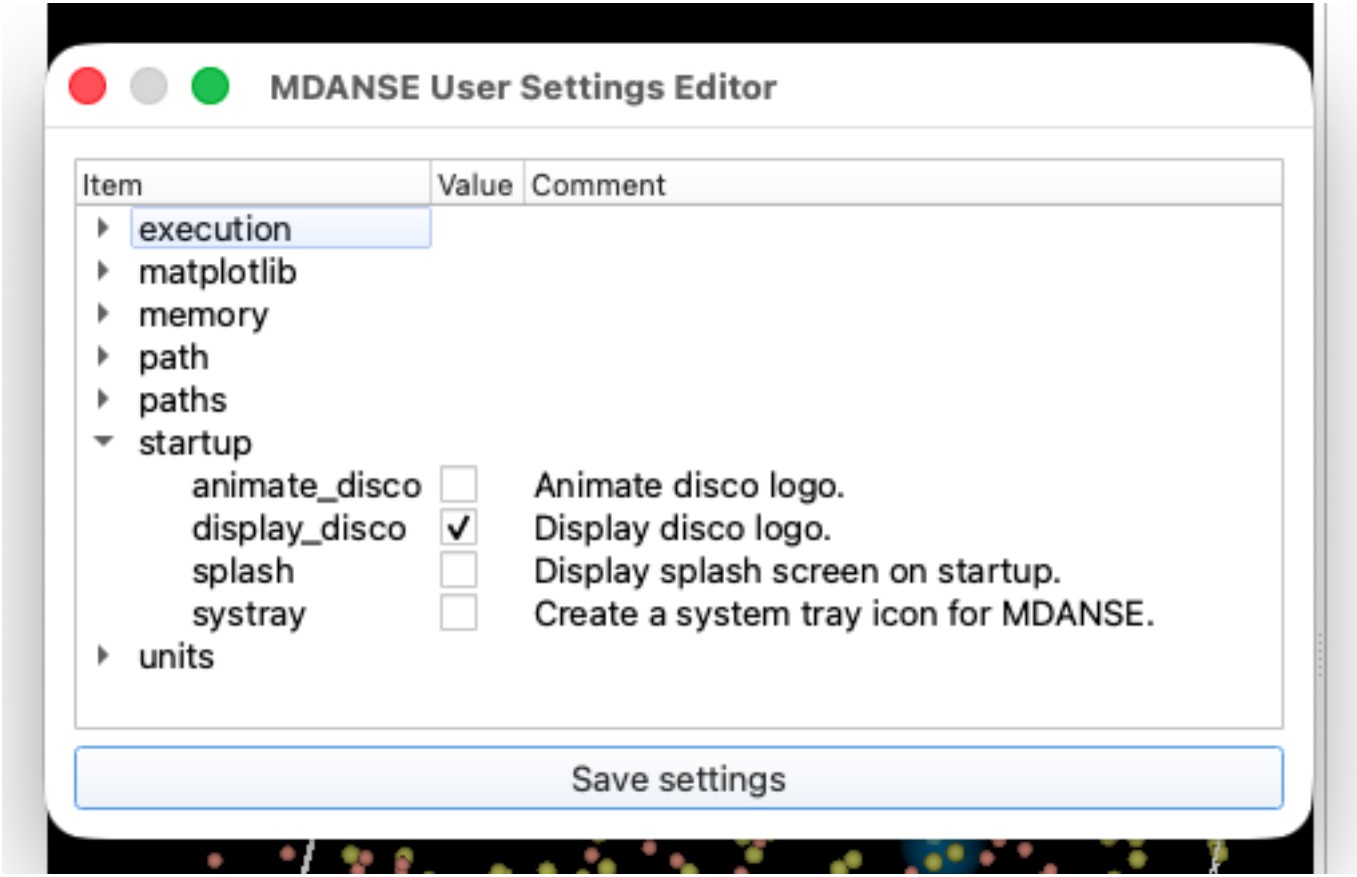
Was it difficult to close the GUI?

Were you not happy with the splash screen?

Did you find the 3D placeholder distracting?

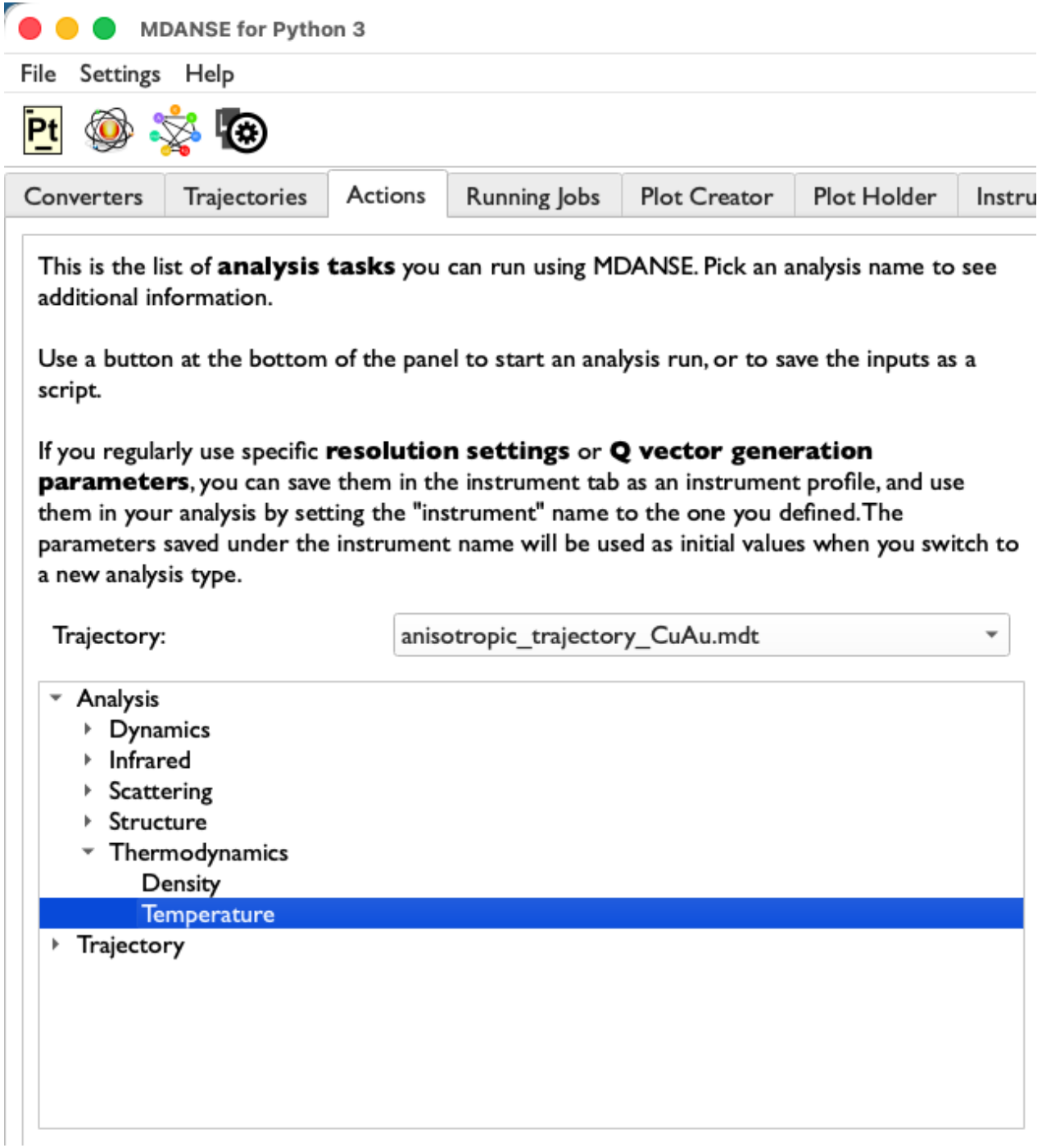
Don’t just tolerate it!

Do something about it!



MDANSE Basics: plotting

Calculate and plot the **temperature** for **both** trajectories



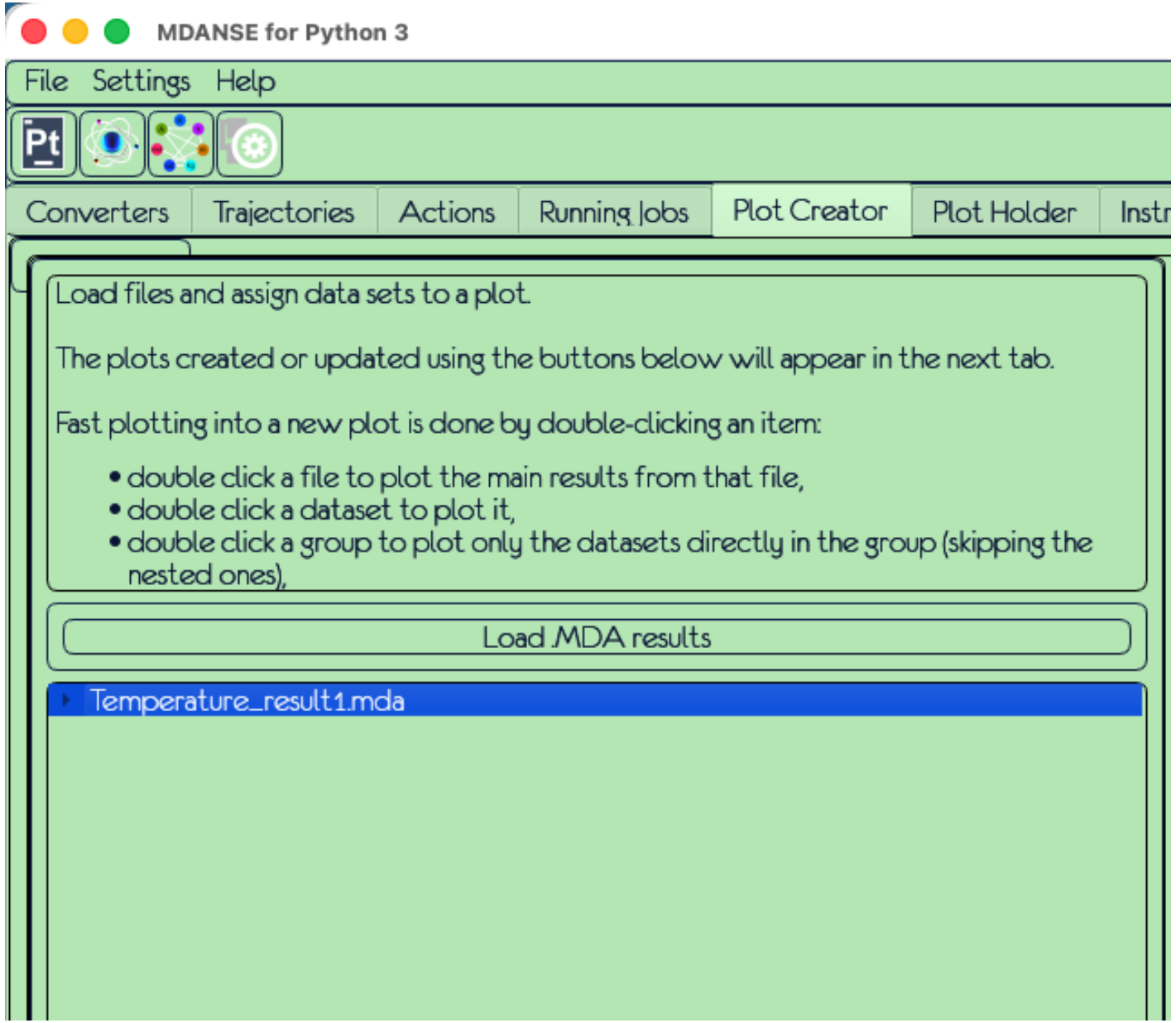
MDANSE Basics: plotting

Calculate and plot the **temperature** for **both** trajectories

Once the calculation is done, try creating a quick plot by **double-clicking the file name** in Plot Creator.

Bonus task:

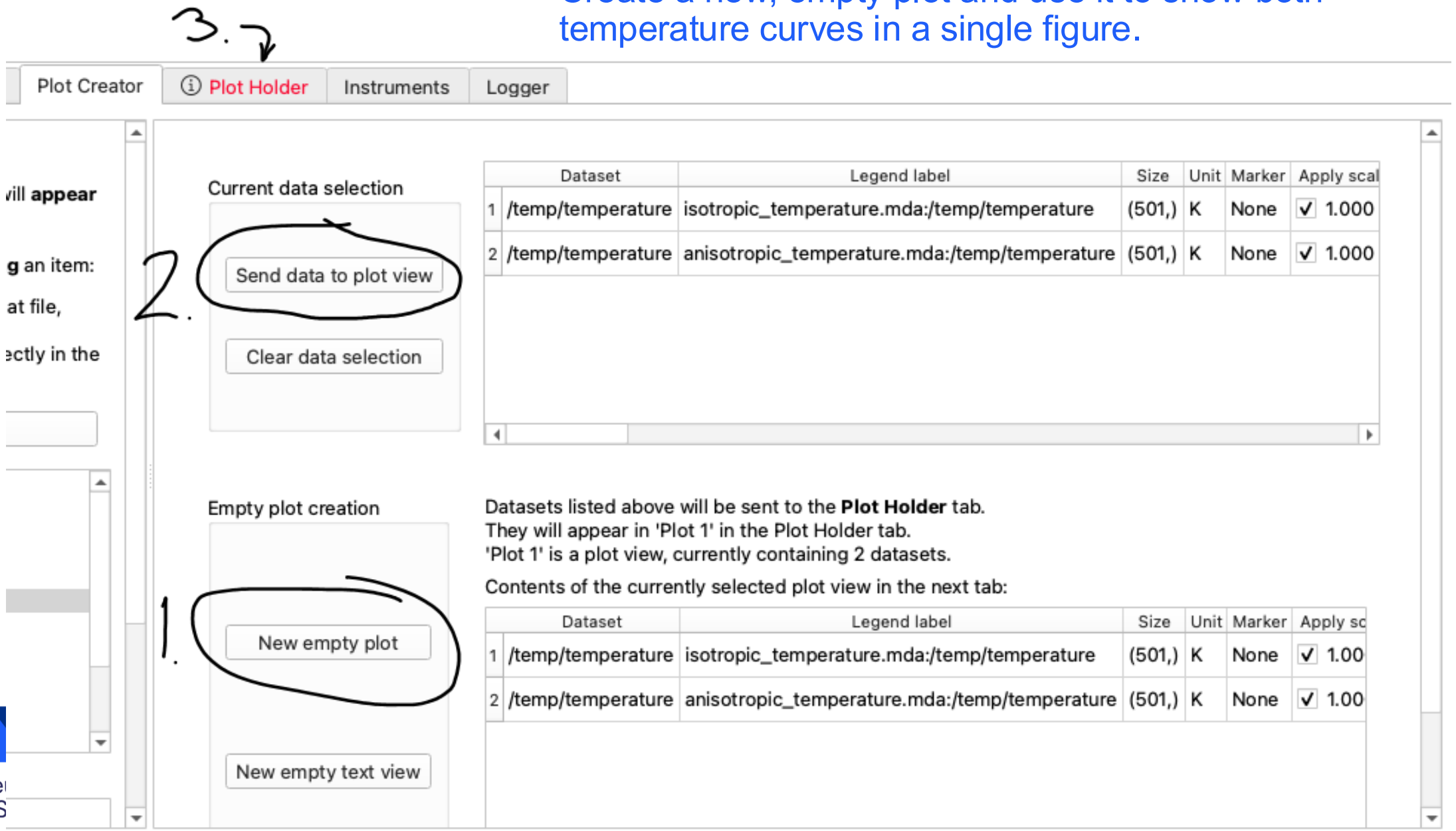
Create a new, empty plot and use it to show both temperature curves in a single figure.



MDANSE Basics: plotting

Bonus task:

Create a new, empty plot and use it to show both temperature curves in a single figure.



MDANSE Basics: scripting

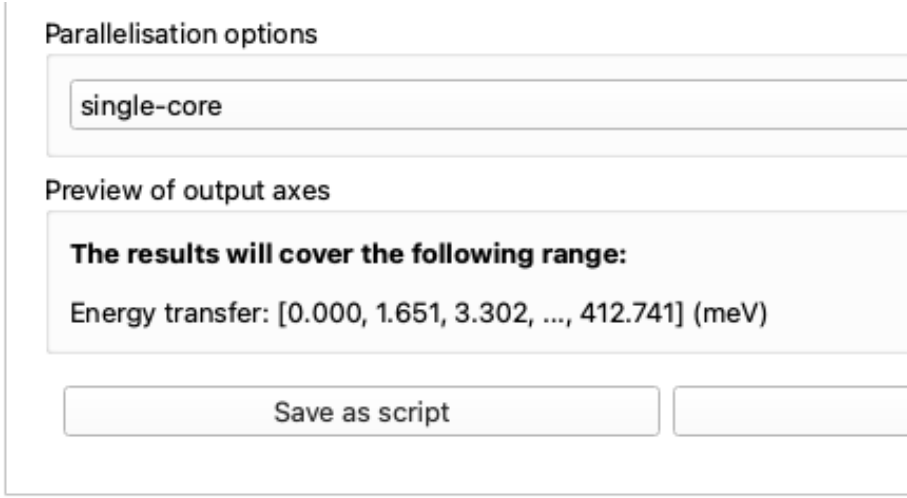
Calculate the vibrational density of states (vDOS) of both trajectories, using axial projection.

We want to calculate the energy spectra of vibrations along specific directions in the system.

We will use projection vectors 100, 010 and 001.

The aim of this task is to use **a script** to perform the calculations.

MDANSE Basics: scripting



We will use projection vectors 100, 010 and 001.

The aim of this task is to use **a script** to perform the calculations.

Steps:

1. In the GUI, go to the Actions tab and select Dynamics->DensityOfStates.
2. Change the **coordinate projection** input to AxialProjection and set the vector to 100.
3. Instead of running the analysis, click “Save as script”. Save the Python script with a meaningful name.
4. Edit the script. Make the script run the analysis 6 times, using 3 different input vectors and 2 different input trajectories.

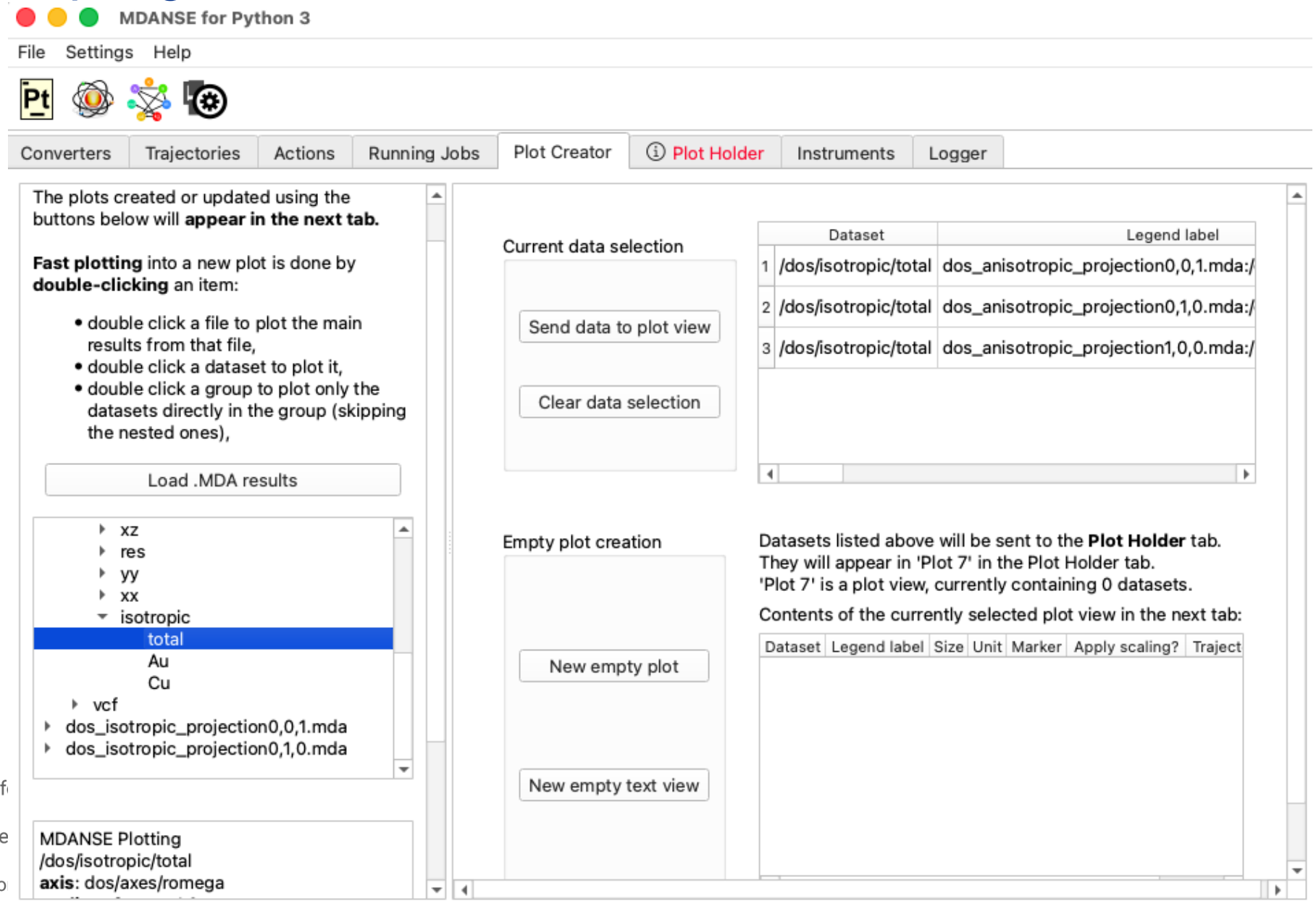
Make sure your script changes the output file name between the analysis runs!

MDANSE Basics: scripting

Run the script. It should create 6 output files.

Load all of them into the GUI in the Plot Creator tab.

Select the total isotropic DOS from different files and send the results to an empty plot.

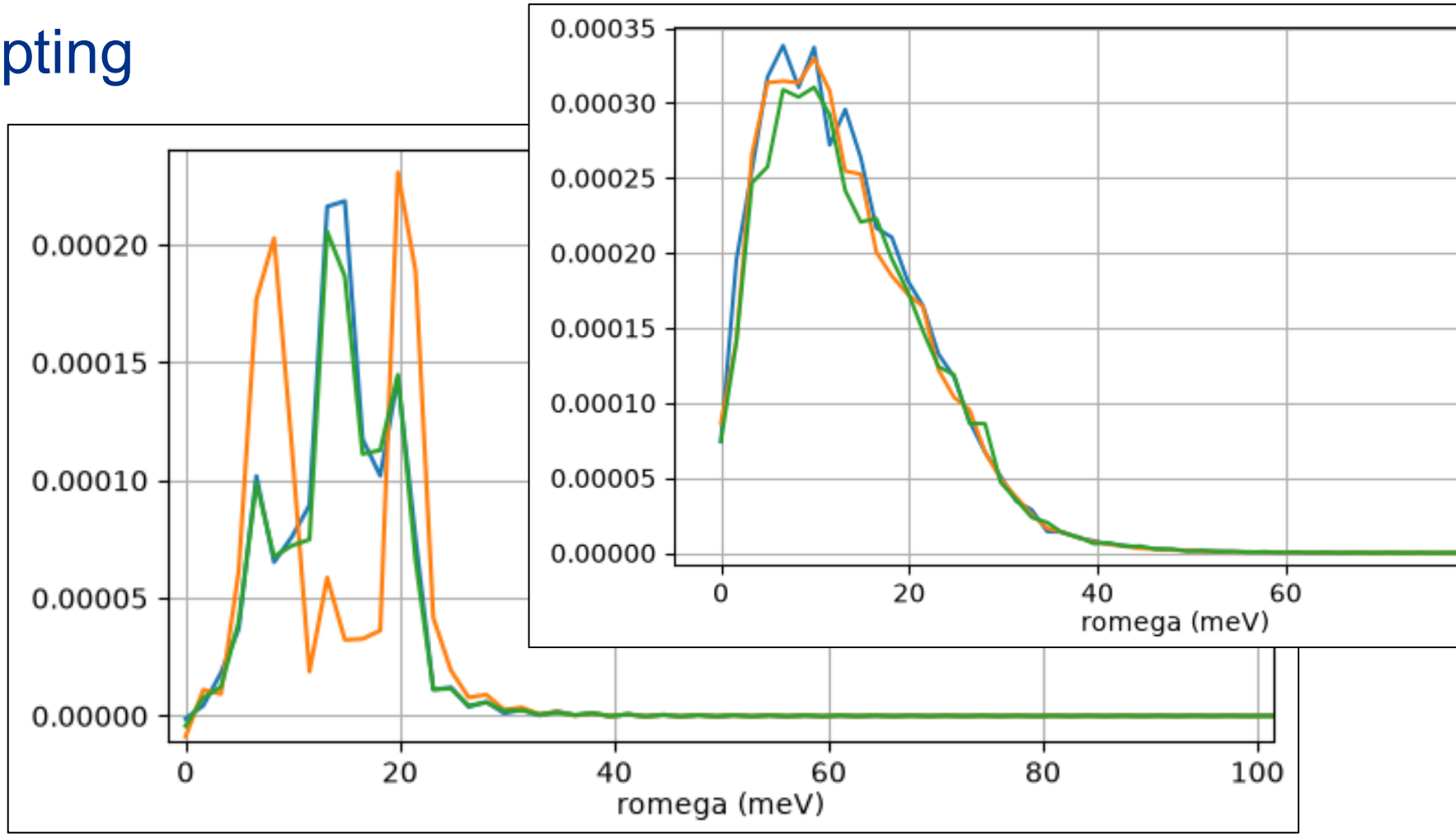


MDANSE Basics: scripting

Run the script. It should create 6 output files.

Load all of them into the GUI in the Plot Creator tab.

For each trajectory, select the total isotropic DOS from different files and send the results to an empty plot.



Did you get plots similar to this one?

Can you explain the difference between the curves?

MDANSE Basics: more plotting

For each trajectory, run the StaticStructureFactor3D analysis.

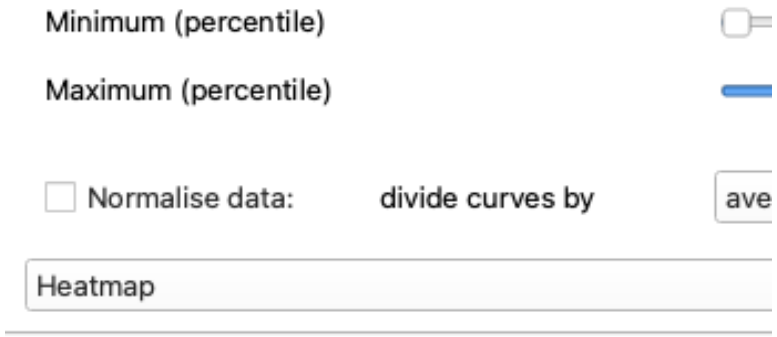
The most important part is the definition of the q-vector grid.

3D Q vector generator

1	seed	0	IntegerConfig...
2	n_samples	100	IntegerConfig...
3	supercell_used	(6, 6, 6)	VectorConfig...
4	q1	(1.0, 0.0, 0.0)	QVectors3DV...
5	q2	(0.0, 1.0, 0.0)	QVectors3DV...
6	q3	q1 x q2	QVectors3DV...
7	q1_range	(-3.0, 3.0, 0.166)	RangeConfig...
8	q2_range	(-3.0, 3.0, ...	RangeConfig...
9	q3_range	(0.0, 0.0, 0.1)	RangeConfig...

MDANSE Basics: more plotting

Plot the results of the SSF3D calculation as Heatmap.



You will have to change the **main axis** of the dataset, and the **index range** used for plotting.

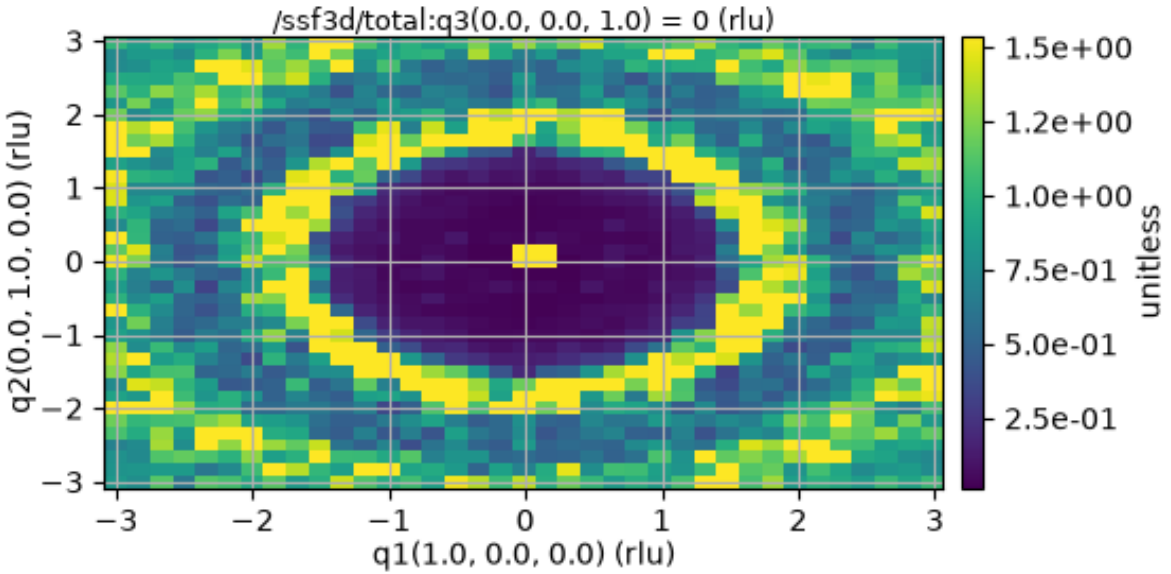
	Legend label	Size	Unit	Main axis	Use it?
3d/total	diffraction_is...	(37, 37, 1)	unitless	q3(0.0, 0.0, 1....	✓ 0
3d/total	diffraction_an...	(37, 37, 1)	unitless	q3(0.0, 0.0, 1....	✓ 0

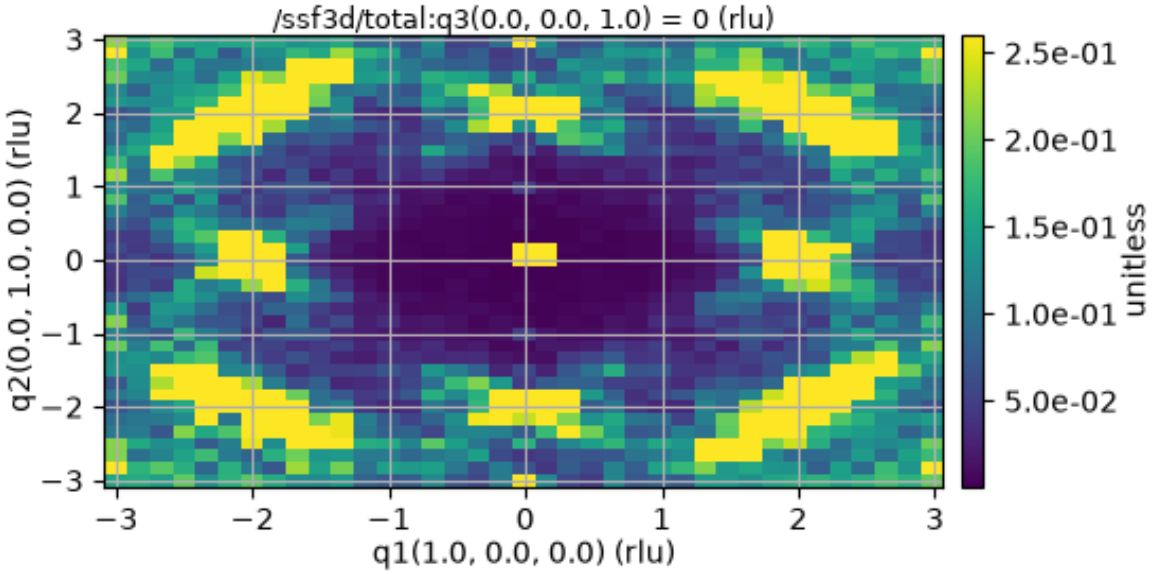
MDANSE Basics: more plotting

Adjust the sliders under the plot.

It will change the limits of the colour map.

Did you get a result like this?





MDANSE Basics: trajectory file structure

Load the “tutorial1_trajectory.h5” file.

This file is in the H5MD format.

https://www.nongnu.org/h5md/h5md.html

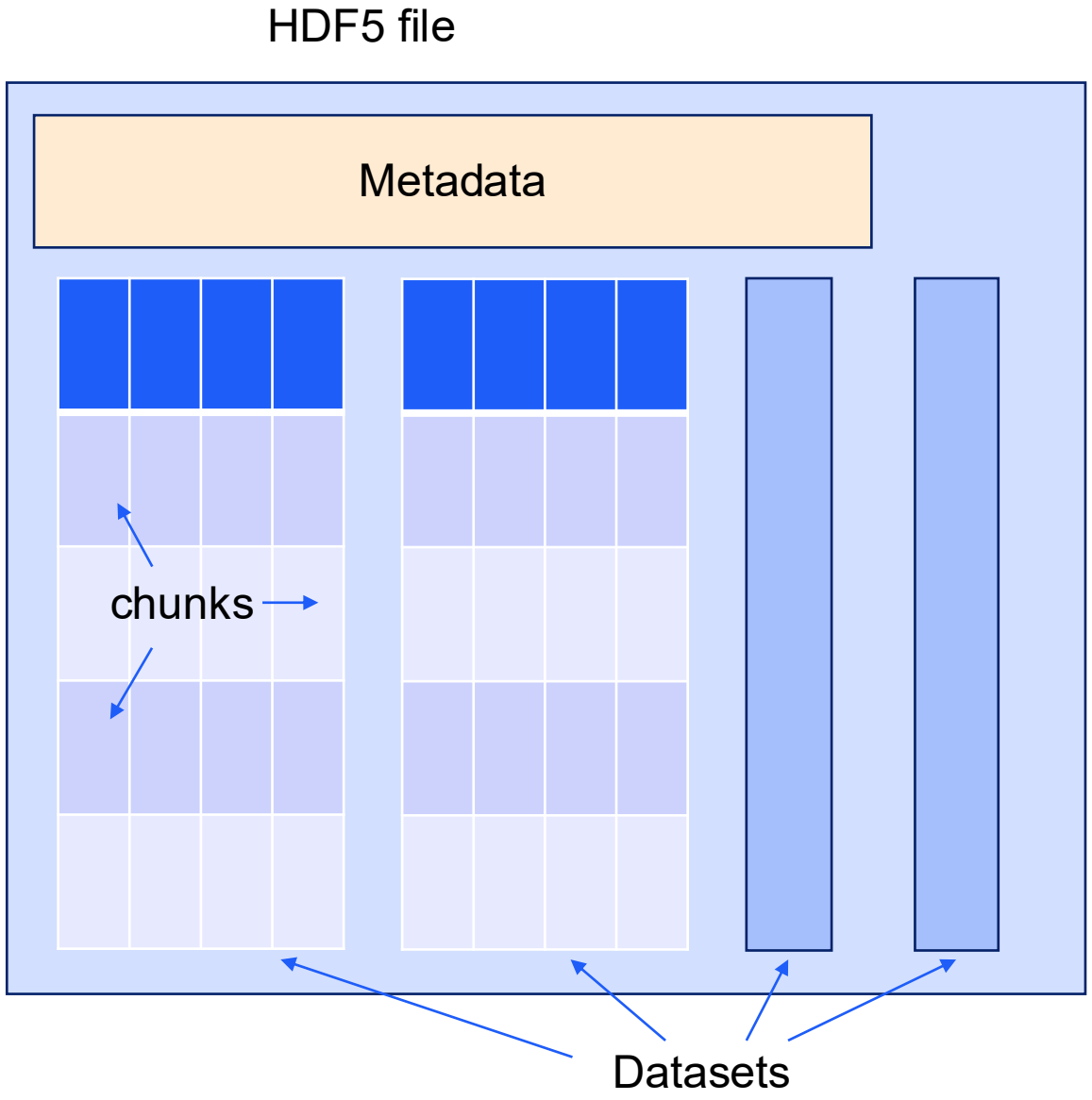
We will run another exercise using this trajectory in a moment.

... **but first, let’s run TrajectoryEditor on it.**

MDANSE Basics: trajectory file structure

Both H5MD and MDT files are, internally, HDF5.

One of their advantages is that it is not necessary to read the entire file.



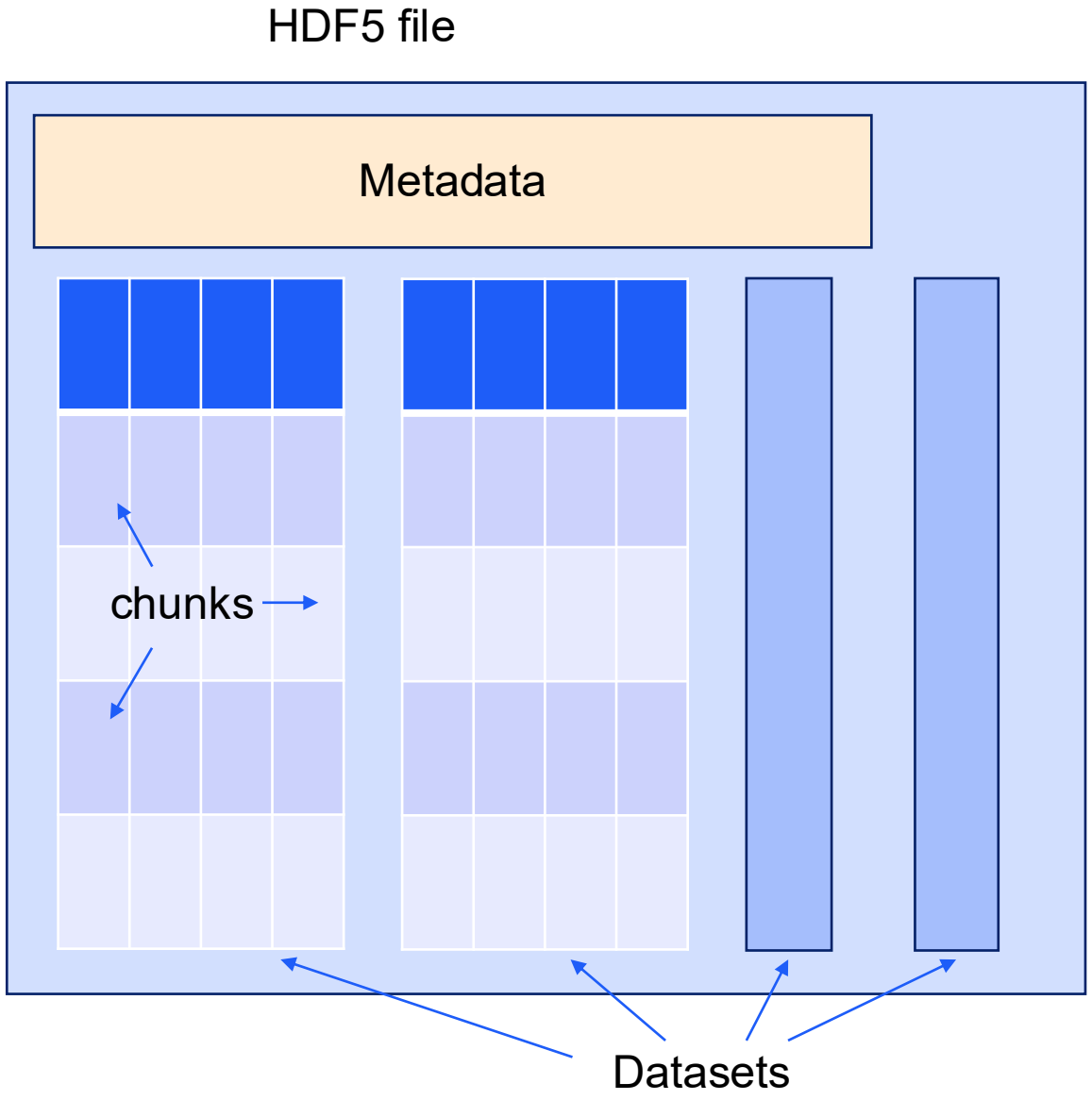
MDANSE Basics: trajectory file structure

We definitely need to read the metadata, to find out what datasets are in the file, and how large they are.

We can then load just the datasets we wanted.

If a dataset is chunked, we can just load a specific chunk.

To get a single value, we still need to read the entire chunk.



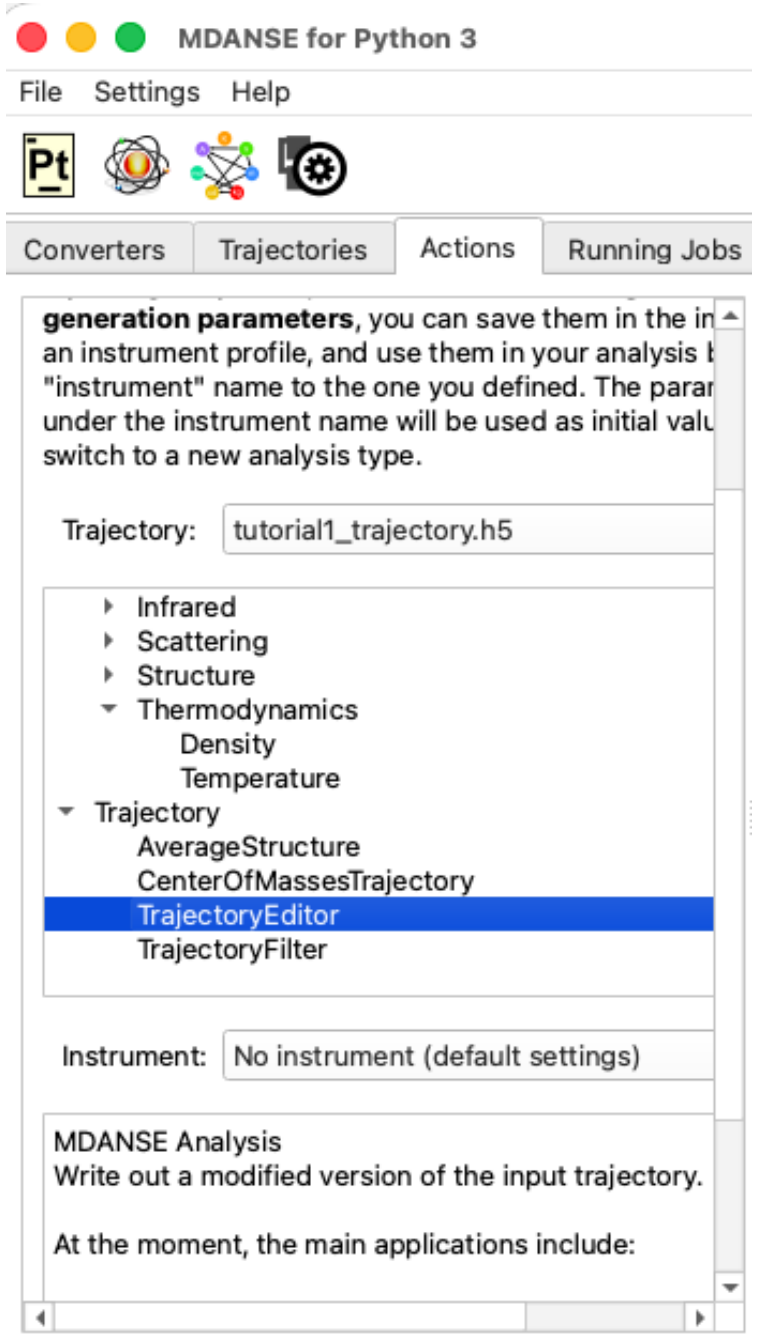
MDANSE Basics: trajectory file structure

To get a single value, we still need to read the entire chunk.

What is the right chunk size?

Try running the TrajectoryEditor a few times, creating new trajectories with:

1. many frames in one chunk,
2. all the atoms in one chunk,
3. generally huge chunks.



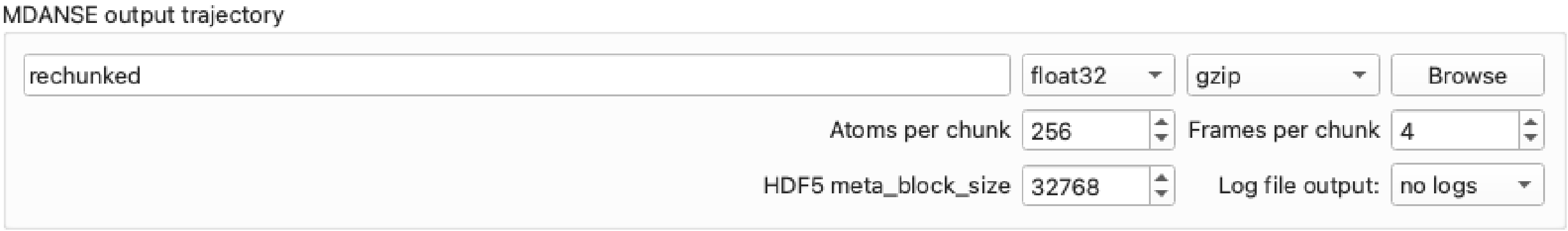
MDANSE Basics: trajectory file structure

To get a single value, we still need to read the entire chunk.

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Try running the TrajectoryEditor a few times, creating new trajectories with:

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MDANSE Basics: trajectory file structure

Try running the TrajectoryEditor a few times, creating new trajectories with:

1. many frames in one chunk,
2. all the atoms in one chunk,
3. generally huge chunks.

Now, compare the files.

- how did the file size change?
- how long does it take to run different analysis types?
  (idea: try Pair Distribution Function **and** Mean Square Displacement)

MDANSE Basics: simulation stability

You've been given a trajectory called "tutorial1_trajectory.h5".

Your task is to **determine if the simulated system is stable**.

Some ideas:

- View the trajectory.
- Run different analysis types.
- Compare analysis results for different frame ranges.

Let us know if you find any problems with this trajectory.

The MDANSE Basics tutorial is now over.

Thank you for participating!